

Anomaly Detection in Ship Engine Functionality

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1. Introduction/Overview of the Problem

The shipping industry relies heavily on the proper functioning of a ship's engines, as malfunctions can lead to safety hazards, operational downtime, and financial losses. This project uses a real dataset to identify anomalous activity in engine functionality (Devabrat, 2022). Typically, anomalies constitute a small proportion of the data, approximately 1–5%. Timely detection allows predictive maintenance, reducing downtime, fuel wastage, and safety risks, ultimately improving operational efficiency and customer satisfaction.

The dataset contains six key features that continuously monitor engine status:

- Engine RPM
- Lubrication oil pressure
- Fuel pressure
- Coolant pressure
- Lubrication oil temperature
- Coolant temperature.

These features collectively indicate whether the engine is operating normally or exhibiting anomalous behaviour.

2. Methodology/Approach

The anomaly detection system in this project was developed using two primary approaches:

- Statistical Methods: Using the Interquartile Range (IQR)
- Unsupervised Machine Learning (ML) methods:
 - o One-Class Support Vector Machine (SVM)
 - o Isolation Forest

2.1 Statistical Method

The IQR method is a classical statistical approach to detect outliers. We first computed the first (Q1) and third quartiles (Q3) for each feature and identified samples with values outside the range of the lower and upper limit as anomalies. Analysis of the results showed that to keep the anomalies within the desired 1–5% range, a sample needed to have at least **two** anomalous features out of **six**, ensuring that only truly unusual observations were flagged.

2.2 Unsupervised Machine Learning Methods

For the unsupervised ML approach, we employed One-Class SVM and Isolation Forest. Both methods do not require labelled anomalies, making them suitable for this problem where anomalies are rare and not predefined.

One-Class SVM constructs a boundary around the normal data points in the feature space and flags points outside this boundary as outliers. Isolation Forest isolates observations by randomly

partitioning the data. Anomalous points tend to require fewer partitions to isolate and are identified as outliers. We initially trained the models with baseline hyperparameters and iteratively tuned them to maintain an anomaly rate within the target 1–5% range. The final configurations for both methods resulted in approximately **3% anomalies**, consistent with expectations.

To visualise the anomalies effectively, Principal Component Analysis (PCA) was applied to reduce the six-dimensional feature space to two components. PCA captures linear relationships among features and projects the data into a lower-dimensional space while preserving most of the variance. This allowed us to plot the data points, distinguishing between normal observations and anomalies. The resulting PCA plots for both One-Class SVM and Isolation Forest revealed a similar distribution of anomalies. Normal data points clustered together, while anomalies were more dispersed and located at the peripheries. This visual consistency indicates that both techniques identified outliers in a comparable manner, reinforcing the robustness of the detection methods.

3. Results & Discussions

Using the IQR statistical method, we identified anomalies by flagging samples that had at least **two** anomalous features out of **six**. This approach resulted in an anomaly rate of **2.16%**, keeping the detected outliers within the desired 1–5% range.

For the One-Class SVM, we set the hyperparameters to **gamma = 0.3** and **nu = 0.03**. After tuning, the model detected approximately **3.02%** of anomalies. The anomalies identified by One-Class SVM were largely consistent with those found using the IQR method, demonstrating alignment between the statistical and machine learning approaches.

The Isolation Forest was configured with 100 estimator and **contamination = 0.03**. This model also detected around **3.005%** of anomalies. The pattern of anomalies identified by Isolation Forest closely resembled that of One-Class SVM, reinforcing the consistency and reliability of the unsupervised machine learning methods in capturing outliers.

Both plots highlight clear separation between normal and anomalous observations, demonstrating the utility of PCA in visualising outliers.

The analysis indicates that:

- Statistical and ML methods provide consistent anomaly detection when tuned appropriately.
- IQR is a simple but effective baseline, particularly useful for establishing an expected anomaly range.
- Unsupervised ML models offer flexibility and can adapt to more complex patterns that might not be captured by simple statistical thresholds.
- PCA-based 2D visualisation is effective for providing intuitive insights into anomaly distribution, though it only captures linear relationships. For highly non-linear structures, alternative methods such as UMAP or t-SNE could be considered.

4. Conclusion

This project successfully developed an anomaly detection framework for monitoring ship engine functionality using both statistical and unsupervised ML approaches. The system reliably identifies approximately 3% of anomalies, aligning with the desired proportion of unusual observations. PCA-based visualisation enabled clear illustration of the separation between normal and anomalous data points, providing an interpretable view for stakeholders. Implementing this framework can help shipping companies reduce downtime, improve safety, optimise maintenance schedules, and enhance overall operational efficiency.

5. Recommendations

Based on the findings:

- Implement a hybrid anomaly detection system: Combine statistical thresholds (IQR) with unsupervised ML to leverage both simplicity and adaptability.
- Automate monitoring and alerting: Set up real-time monitoring of engine parameters with automated alerts when anomalies are detected.
- Periodic hyperparameter tuning: Regularly review and adjust ML model parameters to maintain an anomaly rate within the expected 1–5% range.
- Consider non-linear dimensionality reduction for complex patterns: If anomalies are more subtle or non-linear, tools like t-SNE could improve visual interpretation.
- Integrate predictive maintenance strategies: Use detected anomalies to schedule timely inspections and maintenance, reducing downtime, fuel consumption, and safety risks.